GE Energy

GEK-SA-1040

Mark* VIe Distributed Control System (DCS) Network Design for EMAL

Instruction Guide

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Safety Symbol Legend



Warning

Indicates a procedure, condition, or statement that, if not strictly observed, could result in personal injury or death.



Caution

Indicates a procedure, condition, or statement that, if not strictly observed, could result in damage to or destruction of equipment.



Attention

Indicates a procedure, condition, or statement that should be strictly followed to improve these applications.

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Introduction

This document provides information on the design criteria and configuration considerations applied to the communication network at the EMAL site. The following information is discussed:

- DCS network and topology
- Switch hardware
- Configuration directives
- Other network technologies

Definitions

Ethernet Global Data (EGD) is the control network and protocol for the controller. Devices share data through EGD exchanges (pages).

Gas Turbine (GT) is a mechanical device that converts fuel into rotational energy. This rotation is used for turning an electric generator.

Heat Recovery Steam Generator (HRSG) is a heat exchanger that converts the exhaust of the GT into usable steam.

Plant Data Highway (PDH) is a network that carries plant-wide traffic only.

Rapid Spanning Tree Protocol (RSTP) is a method of monitoring changes to the network topology. It searches for *loops* in the star topology and attempts to block them. These *loops* are the main cause of broadcast storms on a network.

Simple Network Monitoring Protocol (SNMP) is a method of querying a network device and obtaining basic information about the device. This can be used to remotely monitor the health of specific components on a network.

Unit Data Highway (UDH) is a network that carries process-level traffic only. The UDH is used when a GT communicates with a HRSG.

Virtual Local Area Network (VLAN) is a method of allowing different network traffic to operate independently on a single switch. This is used in DCS networks to allow UDH and PDH traffic to operate on a single switch without interacting.

Initial Design Requirements

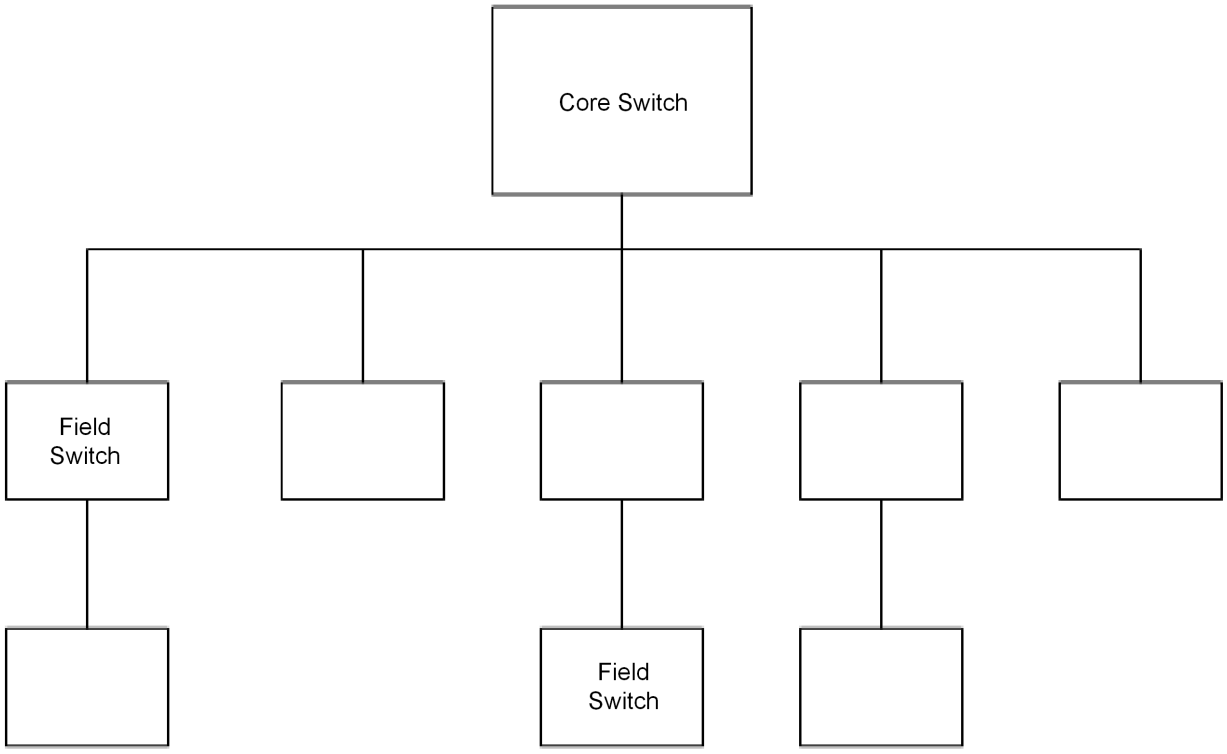
The purpose of a communications network is to allow communications traffic to propagate through the network, and allow specific packets to arrive at their designated destinations. The following criteria of this communications network dictates the design of the network:

- **Population** – In order to size the network, it is necessary to know how many controllers and other devices will be producing data. This population will dictate certain parameters used to configure the size of the network.
- **Protocols** – It is necessary to know what communication protocols the network will be carrying, which dictates some of the design choices for the network. The primary protocol is EGD. Another protocol used is User Datagram Protocol (UDP).

Network Design

Star Network Topology

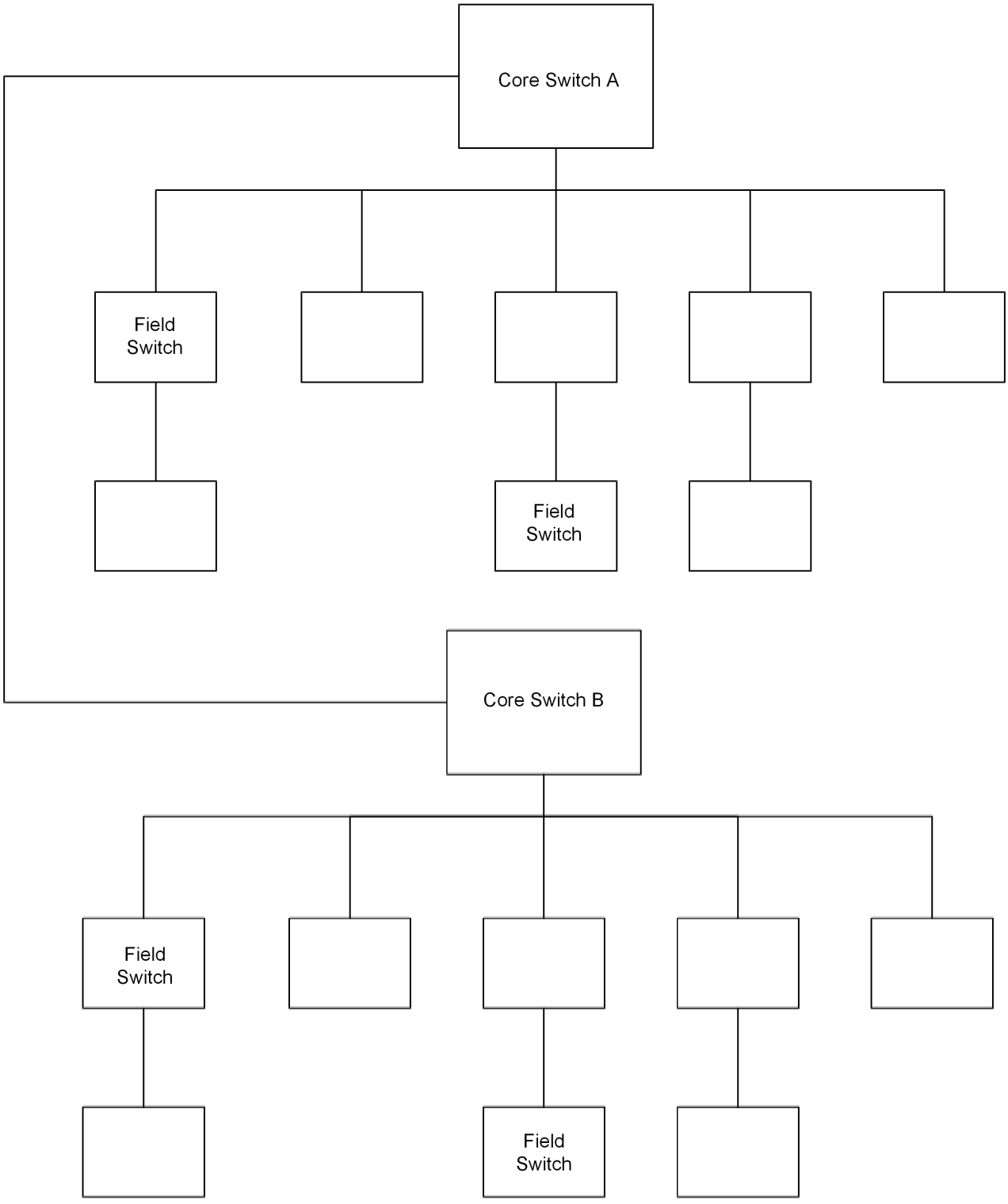
The star topology is a basic hierarchical design, with large-capacity network switches at the *core*, and *field* switches branching off from the core. In the following example, the network is three levels deep. This network depth impacts communication propagation time. This topology, combined with a broadcast communications protocol, ensures timely propagation of information through the network.



Single Star Network

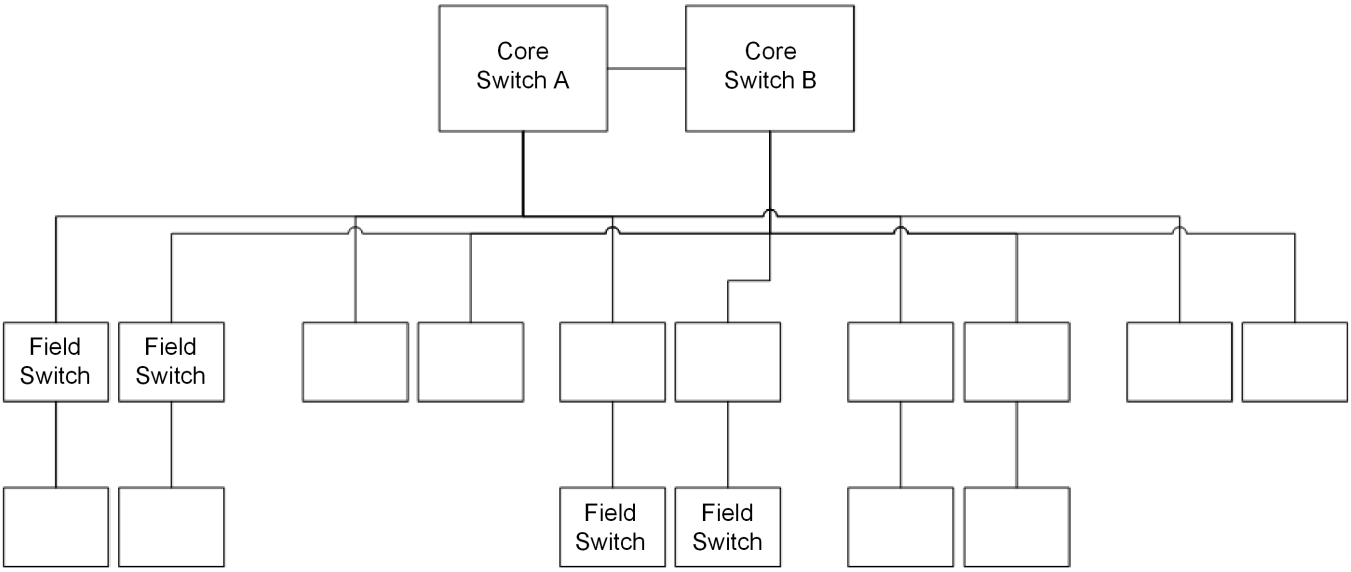
Redundancy

The star network topology has no inherent redundancy. In order to achieve redundancy, a second star network is used in parallel with the first network. This second network only has one connection to the first network through the core switches. The two networks look identical in layout, as displayed in the following figure.



Parallel Redundant Star Networks

In the following example, two networks are physically located together, resulting in a meshed, star topology network.



Meshed Star Networks

In this configuration, there is a pair of network switches at each location. Redundancy is achieved for specific subsystems in the following ways:

- One connection to each network on a single device
- Multiple devices with at least one connection to each network

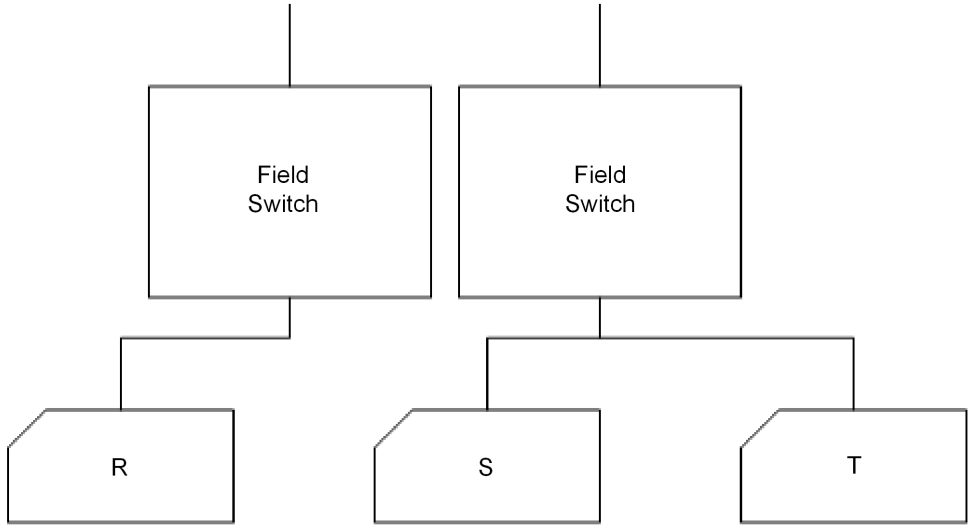
Network Examples

HMIs use four network connections:

- Primary PDH network connection to network A
- Secondary PDH network connection to network B
- Primary UDH network connection to network A
- Secondary UDH network connection to network B

Controllers use one UDH network connection, but redundancy is achieved by having multiple controllers. The following connections are used to achieve triple redundant (TMR) controllers:

- R controller is connected to the UDH port on network A
- S controller is connected to the UDH port on network B
- T controller is connected to the UDH port on network B



Note Controllers have separate IONets for communication to the distributed I/O modules. Controllers do not connect to the PDH.

Switch Configuration

There are three main switches used on a DCS network:

- One Enterasys N-Series network switch
- Two types of Allied Telesyn Network switches

Enterasys N-Series Switch

The Enterasys N-Series switch is a blade-style network switch. This switch offers many features along with the ability to handle a massive amount of network traffic. These switches are used at the core of large DCS networks.

The configuration file used for the Enterasys is a text file, containing the directives and settings. Any unused features are explicitly disabled. The configuration file is organized into several sections:

- **Router** – This configuration covers the IP addresses of the blade switch itself. These directives determine how to connect the various subnets that travel through this switch.
- **Basic Switch** – This section configures generic switch features, such as the IP address of the switch itself (used for administration) and the banner displayed when logged into the switch. The long list of configuration directives either default to disabled or are explicitly disabled for a particular feature.
- **Policy** – This section contains a long list of network rules that allow or deny certain kinds of traffic on the network.

For example, the following configuration statement creates a rule in set 13 that *forwards* (or allows) UDP port 53 traffic on the network. This particular rule is important since this port is used by the Domain Name Service (DNS).

```
set policy rule 13 udpdestportIP 53 mask 16 forward
```

There are dozens of these directives, one for each port, protocol, or address that need to be explicitly allowed, or blocked. Included in the Policy section is a list of all the ports used on the blade switch and their broadcast limits (in packets per second).

- **Simple Network Monitoring Protocol (SNMP)** is a method of querying a network device and obtaining basic information about the device. SNMP can be used to remotely monitor the health of specific components on a network.

This section configures the SNMP features used on the blade switch. These specifically enable read-only and read-write passwords to limit who can access this service. The WorkstationST* Network Monitor uses this service to monitor the health of the blade switch.

- **Rapid Spanning Tree Protocol (RSTP)** is a method of monitoring changes to the network topology. It searches for *loops* in the star topology and attempts to block them. These *loops* are the main cause of broadcast storms on a network.

A switch takes traffic that comes into it and, if it meets certain criteria, delivers it either to a directly connected device or forwards it along. When a loop is present, any forwarded packets get duplicated. This duplication results in an exponential growth of network traffic that will consume all available network bandwidth. The end result is a broadcast storm in which normal network traffic is inhibited by the influx of duplicate packets.

The usual broadcast storm defense is to limit the amount of broadcast packets that a switch can handle before *dropping* any further packets. Although this defense works to keep the switch operating, there is no way to know if a particular dropped packet is part of the broadcast storm (noise) or if it is a real packet (signal). RSTP is a means to prevent the loops that produce broadcast storms in the first place. However, it does take time to calculate necessary changes to the network.

- The **VLAN** (or **Virtual LAN**) configuration directives are scattered throughout the file. At the end of the configuration file resides a VLAN section that consolidates the scattered configurations. It links the policy rules, port assignments, and various other directives to specific VLANs, and assigns a name to each VLAN.

Allied Telesyn AT8516

Allied Telesyn is not a GE entity.

The AT8516 is a 16-port fiber switch. It has minimal configuration file, but it must be configured to match its network.

This config file is a simple text file. Specific directives are only mentioned if they default to disabled, and forced off if they are not required. Although there are no specific sections, the config can be split into the following categories:

- **General Config** – The general configuration for the Allied Telesyn switches is scattered throughout the file. These disable unused features, and configure necessary features.

- **Policy** – This switch has rudimentary policy filters. The main use of the policy is to set the broadcast storm packet limit.

- **VLAN** – Since these switches operate as network aggregators, all configured ports carry the main VLANs (UDH or PDH). Although the traffic is still separate, it operates on the same switch.

- **RSTP** is required to be enabled on all managed switches on the network. This section enables it for all ports – used or otherwise.

- **SNMP** is required in order to monitor switch health. This switch has limited SNMP features, but all of them are made available to the WorkstationST* Network Monitor.

Allied Telesyn AT8624

Allied Telesyn is not a GE entity.

The AT8624 is a 24-port copper switch with two fiber uplink ports. This switch is the workhorse of the DCS network, acting as the field switches (displayed in previous diagrams).

This config file is a simple text file. Specific directives are only mentioned if they default to disabled, and forced off if they are not required. Although there are no specific sections, the config can be split into the following categories:

- **General Config** – The general configuration for the Allied Telesyn switches is scattered throughout the file. These disable unused features, and configure necessary features.
- **Policy** – This switch has rudimentary policy filters. The main use of the policy is to set the broadcast storm packet limit.
- **VLAN** – Since these switches operate as network endpoints, all specific ports are configured to carry specific VLANs (UDH or PDH). The traffic is still separate, but operates on the same switch.
- **RSTP** is required to be enabled on all managed switches on the network. This section enables it for all ports – used or otherwise.
- **SNMP** is required in order to monitor switch health. This switch has limited SNMP features, but all of them are made available to the WorkstationST Network Monitor.

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